

DataBase Query

```
MySQL
CREATE DATABASE shop_management;

USE shop_management;

-- 1. USERS (Authentication)
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(50) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL
);

-- 2. CATEGORIES
CREATE TABLE categories (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL UNIQUE
);

-- 3. SUPPLIERS
CREATE TABLE suppliers (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  phone VARCHAR(20)
);

-- 4. CUSTOMERS
CREATE TABLE customers (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  phone VARCHAR(20)
);

-- 5. PRODUCTS
CREATE TABLE products (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  category_id INT,
  supplier_id INT,
  buying_price DECIMAL(10,2),
  extra_cost DECIMAL(10,2),
  selling_price DECIMAL(10,2) NOT NULL,
  quantity INT DEFAULT 0,

  FOREIGN KEY (category_id) REFERENCES categories(id),
  FOREIGN KEY (supplier_id) REFERENCES suppliers(id)
);

/* Note:
quantity = current stock (updated by transactions)
cost fields = for profit calculation later */

-- 6. PURCHASES
CREATE TABLE purchases (
  id INT AUTO_INCREMENT PRIMARY KEY,
  supplier_id INT NOT NULL,
```

```

total_amount DECIMAL(10,2),
is_credit BOOLEAN DEFAULT FALSE,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

FOREIGN KEY (supplier_id) REFERENCES suppliers(id)
);

-- 7. PURCHASE_ITEMS
CREATE TABLE purchase_items (
  id INT AUTO_INCREMENT PRIMARY KEY,
  purchase_id INT,
  product_id INT,
  quantity INT NOT NULL,
  unit_cost DECIMAL(10,2) NOT NULL,

  FOREIGN KEY (purchase_id) REFERENCES purchases(id) ON DELETE CASCADE,
  FOREIGN KEY (product_id) REFERENCES products(id)
);
-- ON DELETE CASCADE = if purchase deleted → its items deleted automatically

-- 8. PAYABLES
CREATE TABLE payables (
  id INT AUTO_INCREMENT PRIMARY KEY,
  supplier_id INT,
  purchase_id INT,
  amount_due DECIMAL(10,2),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  FOREIGN KEY (supplier_id) REFERENCES suppliers(id),
  FOREIGN KEY (purchase_id) REFERENCES purchases(id)
);

-- 9. PAYABLE PAYMENTS
CREATE TABLE payable_payments (
  id INT AUTO_INCREMENT PRIMARY KEY,
  payable_id INT,
  amount_paid DECIMAL(10,2),
  payment_link VARCHAR(255),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  FOREIGN KEY (payable_id) REFERENCES payables(id) ON DELETE CASCADE
);

-- 10. SALES
CREATE TABLE sales (
  id INT AUTO_INCREMENT PRIMARY KEY,
  customer_id INT,
  total_amount DECIMAL(10,2),
  is_credit BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  FOREIGN KEY (customer_id) REFERENCES customers(id)
);

-- 11. SALE_ITEMS
CREATE TABLE sale_items (

```

```
id INT AUTO_INCREMENT PRIMARY KEY,
sale_id INT,
product_id INT,
quantity INT NOT NULL,
unit_price DECIMAL(10,2) NOT NULL,

FOREIGN KEY (sale_id) REFERENCES sales(id) ON DELETE CASCADE,
FOREIGN KEY (product_id) REFERENCES products(id)
);
```

-- 12. RECEIVABLES

```
CREATE TABLE receivables (
id INT AUTO_INCREMENT PRIMARY KEY,
customer_id INT,
sale_id INT,
amount_due DECIMAL(10,2),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

FOREIGN KEY (customer_id) REFERENCES customers(id),
FOREIGN KEY (sale_id) REFERENCES sales(id)
);
```

-- 13. RECEIVABLE PAYMENTS

```
CREATE TABLE receivable_payments (
id INT AUTO_INCREMENT PRIMARY KEY,
receivable_id INT,
amount_paid DECIMAL(10,2),
payment_link VARCHAR(255),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

FOREIGN KEY (receivable_id) REFERENCES receivables(id) ON DELETE CASCADE
);
```

-- Check all tables if registered
SHOW TABLES;

-- Check the detailed structure of the product table
DESCRIBE products;

-- የ ሚሊ ውት እ ዘ ዝ purchase_items የ ተባለ ውን ሰን ጠረዥ (Table) ለ መፍጠር የ ሚያስፈልጉ ውን መላ
የ SQL ኮድ ያ ወጣል ::

```
SHOW CREATE TABLE purchase_items;
```

-- checking the tables correctness
INSERT INTO categories (name) VALUES ('Clothes');

```
INSERT INTO suppliers (name, phone)
VALUES ('Supplier A', '0912345678');
```

```
INSERT INTO products (name, category_id, supplier_id, selling_price, quantity)
VALUES ('T-Shirt', 1, 1, 200, 10);
```

```
SELECT * FROM products;
```